

Tang Sui

Ph.D. Student in Geography | University of Wisconsin–Madison

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RESEARCH PROFILE

Geospatial artificial intelligence researcher developing Earth-observation methods for environmental hazards, with a primary focus on active-fire detection, wildfire-progression forecasting, and post-fire burned-area and damage assessment. Expertise includes multimodal satellite data, spatiotemporal graph neural networks, attention and transformer architectures, remote-sensing reconstruction, and physically informed environmental modeling.

EDUCATION

University of Wisconsin–Madison — Madison, WI

Ph.D. in Geography, May 2026–Present

M.S. in Geography (Cartography/GIS, thesis), Aug. 2024–May 2026; GPA: 4.0/4.0

Tongji University — Shanghai, China

B.Eng. in Surveying and Mapping, Sept. 2019–July 2024

RESEARCH EXPERIENCE

Graduate Research Assistant, Spatial Computing and Data Mining Lab — UW–Madison | Aug. 2024–Present

Develop GeoAI methods for wildfire monitoring, environmental time-series modeling, hyperspectral reconstruction, and remote-sensing super-resolution using satellite, weather, terrain, land-cover, and sensor-network data.

Wildfire monitoring: Built models spanning active-fire detection, multi-step fire-front progression forecasting, and post-fire mapping using multi-sensor imagery and environmental variables.

BiAU-Net: Developed a bi-temporal, attention-enhanced U-Net for burned-area mapping; evaluated generalization across five continents and released data and code.

HydroTGNN: Modeled Iowa stream gauges as a directed river graph and combined spatial graph learning with temporal convolution for hourly streamflow imputation and prediction.

Hyperspectral reconstruction: Developed transformer-based methods for generating hyperspectral imagery from sparse multispectral observations, including 3D shifted-window attention and cascade training.

SELECTED PEER-REVIEWED PUBLICATIONS

Tang Sui, Songxi Yang, and Qunying Huang. “SpecSwin3D: Generating Hyperspectral Imagery From Multispectral Data via Transformer Networks and Curriculum-Based Cascade Training.” *IEEE Transactions on Geoscience and Remote Sensing*, 64, 2026. DOI: 10.1109/TGRS.2026.3662323.

Tang Sui, Cheng Wang, Yi Zhuang, Songxi Yang, Jichao Fang, and Qunying Huang. “HydroTGNN: Filling Missing Streamflow Data with Graph Neural Network.” *Environmental Modelling & Software*, 199:106937, 2026. DOI: 10.1016/j.envsoft.2026.106937.

Songxi Yang, **Tang Sui**, and Qunying Huang. “GeoAwareDiffSR: Geospatial-Aware Remote Sensing Super-Resolution with Diffusion Priors and Multi-Modal Constraints for Crop Mapping.” *GIScience & Remote Sensing*, 62(1):2597564, 2025.

Tang Sui, Songxi Yang, and Qunying Huang. “SpecSwin: Enhancing Hyperspectral Data Generation through Deep Learning and Multispectral Imagery.” *Proceedings of GeoSim 2025*, pp. 85–93. DOI: 10.1145/3764921.3770156.

Tang Sui, Qunying Huang, Mingda Wu, Meiliu Wu, and Zhou Zhang. “BiAU-Net: Wildfire Burnt Area Mapping Using Bi-Temporal Sentinel-2 Imagery and U-Net with Attention Mechanism.” *International Journal of Applied Earth Observation and Geoinformation*, 132:104034, 2024. DOI: 10.1016/j.jag.2024.104034.

SELECTED PEER-REVIEWED PUBLICATIONS — CONTINUED

Songxi Yang, Bo Peng, **Tang Sui**, Meiliu Wu, and Qunying Huang. “Advancing Self-Supervised Learning for Building Change Detection and Damage Assessment.” *Remote Sensing*, 17(15):2717, 2025. DOI: 10.3390/rs17152717.

Mingda Wu, **Tang Sui**, Bo Peng, Manzhu Yu, and Qunying Huang. “A Remote Sensing Spectral Index Guided Bitemporal Residual Attention Network for Wildfire Burn Severity Mapping.” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 17:17187–17206, 2024. DOI: 10.1109/JSTARS.2024.3460531.

Songxi Yang, Bo Peng, **Tang Sui**, and Qunying Huang. “Self-Supervised Pretraining with Edge Guidance for Building Damage Assessment.” *Proceedings of AI4Geo 2024*. DOI: 10.1145/3687123.369828.

HONORS AND AWARDS

Best Student Paper Award / 1st Place, Cartography and Geographic Information Society (CaGIS) Student Paper Competition, 2026 UCGIS Symposium.

UCGIS Student Paper Competition Finalist and Travel Award, 2026.

Finalist, AAG Remote Sensing Specialty Group Student Illustrated Paper Competition, 2026.

Student Research Grants Competition – Conference Presentation Funds, UW–Madison Graduate School, 2026.

Summer Geography Graduate Scholarship, UW–Madison Department of Geography, 2026.

First Prize, National UAV Photogrammetry Virtual Simulation Competition, 2022.

ACADEMIC LEADERSHIP AND PROFESSIONAL SERVICE

Primary organizer and chair, “GeoAI and Deep Learning Symposium: AI for Earth Observation,” AAG Annual Meeting, 2026.

Co-organizer, “The Frontiers of GeoAI,” Symposium on Spatiotemporal Data Science, 2026; presented “GeoAI for Earth Observation and Wildfire Management.”

Peer review: Ad hoc reviewer for Remote Sensing of Environment, Bioresource Technology, and Remote Sensing Applications: Society and Environment.

TEACHING EXPERIENCE

Teaching Assistant, GEOG 578: GIS Applications, UW–Madison, Fall 2025 and Fall 2026.

Teaching Assistant, GEOG 573: Advanced Geocomputing, UW–Madison, Spring 2026.

Grader / instructional support, GEOG 170: Our Digital Globe, UW–Madison, Spring 2025.

TECHNICAL EXPERTISE

Methods: GeoAI, deep learning, CNN/U-Net, transformers, GNNs, ConvLSTM, attention, physics-informed modeling, spatiotemporal analysis. **Data & tools:** Python, PyTorch, TensorFlow, Google Earth Engine, ArcGIS, QGIS; Sentinel-2, VIIRS, GOES, weather, terrain, land-cover, stream-gauge, and hyperspectral data.